

Biquaternions

Hamilton defined the quaternion basis i, j, k like

$$i^2 = j^2 = k^2 = ijk = -1$$

and quaternions to be of the form $q = q_0 + q_1i + q_2j + q_3k$. If we let each q_i be a complex number of the form $a + bh$ with $h^2 = -1$ and commuting with i, j, k , we arrive at the biquaternions (also called the complex quaternions), $\mathbb{Cl}_3^+$. As it turns out the biquaternions are exactly equivalent to the geometric algebra $\text{Cl}_{3,0}$.

$\text{Cl}_{3,0}$ is generated by the basis vectors $\sigma_1, \sigma_2, \sigma_3$ with $\sigma_i^2 = 1$ and $\sigma_i\sigma_j = -\sigma_j\sigma_i$ for $i \neq j$. We will write σ_{ij} for $\sigma_i\sigma_j$ &c.

We find the following multiplication rules:

$$\begin{aligned}\sigma_{12}^2 &= \sigma_{1212} = -\sigma_{1122} = -1 \\ \sigma_{23}^2 &= \sigma_{2323} = -\sigma_{2233} = -1 \\ \sigma_{31}^2 &= \sigma_{3131} = -\sigma_{3131} = -1 \\ \sigma_{12}\sigma_{23} &= \sigma_{13} = -\sigma_{31} \\ \sigma_{23}\sigma_{31} &= \sigma_{21} = -\sigma_{12} \\ \sigma_{31}\sigma_{12} &= \sigma_{32} = -\sigma_{23} \\ \sigma_{123}^2 &= \sigma_{123123} = \sigma_{112323} = -\sigma_{112233} = -1 \\ \sigma_1\sigma_{123} &= \sigma_{123}\sigma_1 = \sigma_{23} \\ \sigma_2\sigma_{123} &= \sigma_{123}\sigma_2 = \sigma_{31} \\ \sigma_3\sigma_{123} &= \sigma_{123}\sigma_3 = \sigma_{12} \\ \sigma_{23}\sigma_{123} &= \sigma_{123}\sigma_{23} = -\sigma_1 \\ \sigma_{31}\sigma_{123} &= \sigma_{123}\sigma_{31} = -\sigma_2 \\ \sigma_{12}\sigma_{123} &= \sigma_{123}\sigma_{12} = -\sigma_3\end{aligned}$$

We can see that the following identification of biquaternion with $\text{Cl}_{3,0}$ elements gives exactly the same rules while obeying the biquaternion definition:

$$\begin{aligned}i &:= -\sigma_{23} \\ j &:= -\sigma_{31} \\ k &:= -\sigma_{12} \\ h &:= \sigma_{123}\end{aligned}$$

In particular it follows that

$$\begin{aligned}ih &= hi = \sigma_1 \\ jh &= hj = \sigma_2 \\ kh &= hk = \sigma_3\end{aligned}$$

Thus every basis element of $\text{Cl}_{3,0}$ has a counterpart in $\mathbb{Cl}_3^+$.

Spacetime algebra

Dirac's γ -matrices are precisely $\text{Cl}_{1,3}$, also called the Spacetime algebra (STA). Let i, j be 1, 2, 3 and distinct. We have:

$$\gamma_0^2 = 1$$

$$\gamma_i^2 = -1$$

We quickly find that the even subalgebra of this is isomorphic to $\text{Cl}_{3,0}$:

$$\gamma_{i0}^2 = 1$$

$$\gamma_{ij}^2 = -1$$

$$\gamma_{0123}^2 = -1$$

$$\gamma_{12}\gamma_{23} = \gamma_{31} = -\sigma_{31}$$

$$\gamma_{23}\gamma_{31} = \gamma_{12} = -\sigma_{12}$$

$$\gamma_{31}\gamma_{12} = \gamma_{23} = -\sigma_{23}$$

$$\gamma_{10}\gamma_{20} = -\gamma_{12}$$

$$\gamma_{20}\gamma_{30} = -\gamma_{23}$$

$$\gamma_{30}\gamma_{10} = -\gamma_{31}$$

$$\gamma_{i0} = \sigma_i$$

$$\sigma_{123} = \gamma_{10}\gamma_{20}\gamma_{30} = (-\gamma_{01})(-\gamma_{23}) = \gamma_{0123}$$

$\text{Cl}_{1,3}$		$\text{Cl}_{3,0}$		Cl_3^+		Cl_2		e^2
1	0	1	0	1	0	1	0	1
γ_0	1							1
γ_1	1							-1
γ_2	1							-1
γ_3	1							-1
γ_{10}	2	σ_1	1	$i\rho_1$	2i	$i\hat{i}$	1i	1
γ_{20}	2	σ_2	1	$i\rho_2$	2i	$i\hat{j}$	1i	1
γ_{30}	2	σ_3	1	$i\rho_3$	2i	$i\hat{k}$	2i	1
γ_{23}	2	$-\sigma_{23}$	2	ρ_1	2	$\hat{i}$	1	-1
γ_{31}	2	$-\sigma_{31}$	2	ρ_2	2	$\hat{j}$	1	-1
γ_{12}	2	$-\sigma_{12}$	2	ρ_3	2	$\hat{k}$	2	-1
γ_{123}	3							1
γ_{023}	3							-1
γ_{031}	3							-1
γ_{012}	3							-1
γ_{0123}	4	σ_{123}	3	i	0i	i	0i	-1

We can generate $\text{Cl}_{3,1}$ by two-sided multiplication of quaternions, and $\text{Cl}_{1,3}$ by two-sided multiplication of complex quaternions:

	$\text{Cl}_{1,3}$	$\text{Cl}_{3,1}$
1	1 1	1 1
γ_0	1 $i\rho_1$	1 ρ_1
γ_1	ρ_1 $i\rho_2$	ρ_1 ρ_2
γ_2	ρ_2 $i\rho_2$	ρ_2 ρ_2
γ_3	ρ_3 $i\rho_2$	ρ_3 ρ_2
γ_{10}	$-\rho_1$ ρ_3	ρ_1 ρ_3
γ_{20}	$-\rho_2$ ρ_3	ρ_2 ρ_3
γ_{30}	$-\rho_3$ ρ_3	ρ_3 ρ_3
γ_{23}	ρ_1 1	$-\rho_1$ 1
γ_{31}	ρ_2 1	$-\rho_2$ 1
γ_{12}	ρ_3 1	$-\rho_3$ 1
γ_{123}	-1 $i\rho_2$	1 ρ_2
γ_{023}	ρ_1 $i\rho_1$	$-\rho_1$ ρ_1
γ_{031}	ρ_2 $i\rho_1$	$-\rho_2$ ρ_1
γ_{012}	ρ_3 $i\rho_1$	$-\rho_3$ ρ_1
γ_{0123}	-1 ρ_3	-1 ρ_3

	$\text{Cl}_{1,3}$	$\text{Cl}_{3,1}$
1 1	1	1
ρ_1 1	γ_{23}	$-\gamma_{23}$
ρ_2 1	γ_{31}	$-\gamma_{31}$
ρ_3 1	γ_{12}	$-\gamma_{12}$
1 ρ_1	$-i\gamma_0$	γ_0
ρ_1 ρ_1	$-i\gamma_{023}$	$-\gamma_{023}$
ρ_2 ρ_1	$-i\gamma_{031}$	$-\gamma_{031}$
ρ_3 ρ_1	$-i\gamma_{012}$	$-\gamma_{012}$
1 ρ_2	$i\gamma_{123}$	γ_{123}
ρ_1 ρ_2	$-i\gamma_1$	γ_1
ρ_2 ρ_2	$-i\gamma_2$	γ_2
ρ_3 ρ_2	$-i\gamma_3$	γ_3
1 ρ_3	$-\gamma_{0123}$	$-\gamma_{0123}$
ρ_1 ρ_3	$-\gamma_{10}$	γ_{10}
ρ_2 ρ_3	$-\gamma_{20}$	γ_{20}
ρ_3 ρ_3	$-\gamma_{30}$	γ_{30}

	$\text{Cl}_{3,1}$			
	1	ρ_1	ρ_2	ρ_3
1	1	γ_0	γ_{123}	$-\gamma_{0123}$
ρ_1	$-\gamma_{23}$	$-\gamma_{023}$	γ_1	γ_{10}
ρ_2	$-\gamma_{31}$	$-\gamma_{031}$	γ_2	γ_{20}
ρ_3	$-\gamma_{12}$	$-\gamma_{012}$	γ_3	γ_{30}

	$\text{Cl}_{1,3}$			
	1	$i\rho_1$	$i\rho_2$	ρ_3
1	1	γ_0	$-\gamma_{123}$	$-\gamma_{0123}$
ρ_1	γ_{23}	γ_{023}	γ_1	$-\gamma_{10}$
ρ_2	γ_{31}	γ_{031}	γ_2	$-\gamma_{20}$
ρ_3	γ_{12}	γ_{012}	γ_3	$-\gamma_{30}$

	1	i	j	k
1	1	γ_0	γ_{123}	$-\gamma_{0123}$
i	$-\gamma_{23}$	$-\gamma_{023}$	γ_1	γ_{10}
j	$-\gamma_{31}$	$-\gamma_{031}$	γ_2	γ_{20}
k	$-\gamma_{12}$	$-\gamma_{012}$	γ_3	γ_{30}

e.g. $\gamma_1\gamma_{10} = i(i|k)j = (ii)|(kj) = 1|i = \gamma_0$